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B.Sc. in Applied Sciences

STA 1303 – Probability Theory
Tutorial No: 1

1. Define the following statistical terms with examples
 - I. Univariate
 - II. Bivariate
 - III. Multivariate
 - IV. Random Variables
 - V. Joint Probability Distribution
 - VI. Marginal Distributions
2. A gas station has both self-service and full-service islands, each with a single regular unleaded pump with two hoses.
X = The number of houses in use on self-service Island
Y = The number of houses in use on the full-service Island

P (X, Y)		Y (full service)		
X (self-service)		0	1	2
	0	0.1	0.04	0.02
	1	0.08	0.20	0.06
	2	0.06	0.14	0.3

- I. What is $p(x = 2, y = 1)$?
 - II. What is $p(X \leq 1, Y \leq 1)$?
 - III. Find $F(1,0)$
3. A bag filled with three blue balls, two red balls, and three white balls is randomly opened to choose the two balls. If Y1 is the number of blue balls selected and Y2 is the number of red balls selected, assume that the two balls are randomly selected from the eight available.
Find the joint probability function of $f(Y_1, Y_2)$
4. Consider the following joint density function of random variables Y1 and Y2:

$$f(Y_1, Y_2) = \begin{cases} \frac{2}{5}(2Y_1 + 3Y_2), & 0 \leq Y_1 \leq 1, \quad 0 \leq Y_2 \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

- I. Verify that the probability function satisfies the theorem 1
- II. Find the $P[(Y_1, Y_2), (Y_1, Y_2) | 0 < Y_1 < \frac{1}{2}, \frac{1}{4} < Y_2 < \frac{1}{2}]$

5. Let Y_1 and Y_2 have the joint probability density function given by

$$f(Y_1, Y_2) = \begin{cases} K(1 - Y_2) & 0 \leq Y_1 \leq Y_2 \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

Find the value of “K” that makes this a valid probability density function

6. refer to the problem no 3, find the conditional distribution of Y_1 , given that $Y_2 = 1$.
7. The joint density for the random variables (Y_1, Y_2) , where Y_1 is the unit temperature change and Y_2 is proportion spectrum shift that a certain atomic particle produces, is

$$f(Y_1, Y_2) = \begin{cases} 10Y_1Y_2^2, & 0 < Y_1 < Y_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- I. Find the marginal densities $f_1(Y_1)$, $f_2(Y_2)$, and the conditional density $f(Y_2 | Y_1)$
 - II. Find the probability that the spectrum shifts more than half of the total observations, given that the temperature increased by 0.25 unit.
8. consider the following joint probability functions for Y_1 and Y_2 . Are they independent or not? Show the proof.

$$f(Y_1, Y_2) = \begin{cases} Y_1 + Y_2, & 0 < Y_1 < 1, \quad 0 < Y_2 < 1 \\ 0, & \text{elsewhere} \end{cases}$$

9. consider the following joint probability function of Y_1 and Y_2 .

$$f(Y_1, Y_2) = \begin{cases} 4Y_1Y_2, & 0 < Y_1 < 1, \quad 0 < Y_2 < 1 \\ 0 & \text{elsewhere} \end{cases}$$

- I. Check the they independent or not
- II. Find $E(Y_1)$
- III. Find $V(Y_1)$
- IV. Find $E(Y_1 - Y_2)$

10. Using the previous problem, show the $COV(Y_1, Y_2) = 0$, and explain why it is zero.

(Please answer question number 3, 5, 6, 7, 9, and 10 and submit the answer script to the tutorial room on or before 4th of October 2023)